CS 5472 - Advanced Topics in Computer Security

Topic 8: Ransomware (1)

Spring 2018 Semester

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Final Project Presentation

- April 17, April 19, potentially part of April 24. Arranged according to the alphabetical order of students' last names
  - Sandeep Battula
  - Manu Nandan Chemudupati
  - Sophia Farquhar
  - Ali Jalooli
  - Rahul Javadekar
  - Krishna Kakula
  - Shashank Munnooru
  - Ryan Olson
  - Abheek Srivastava
- Each student has 25 minutes (presentation + Q&A)
  - Problem/motivation (what is the problem? Why doing it?)
  - Related work (What have been done?)
  - Solution (what is your solution?)
  - Evaluation (How can you convince us that your solution is good?)

Final Project Presentation (cont.)

- Survey: Do you want to reduce one more summary so that you can have more time to finish your project? (I observed the quality of summary is improved)

Ransomware Attacks Keep Growing …

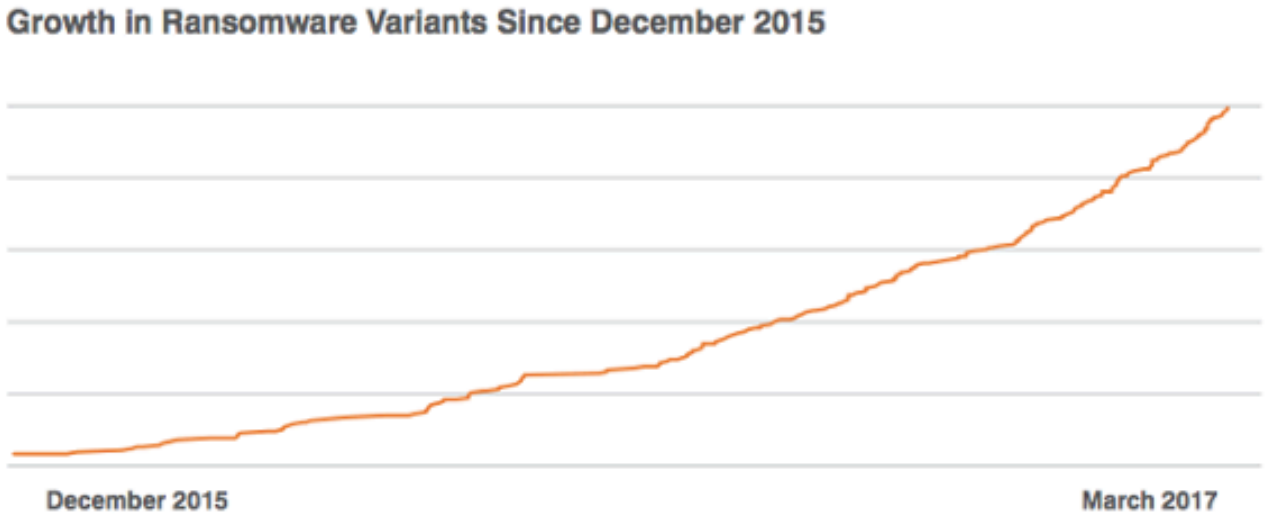
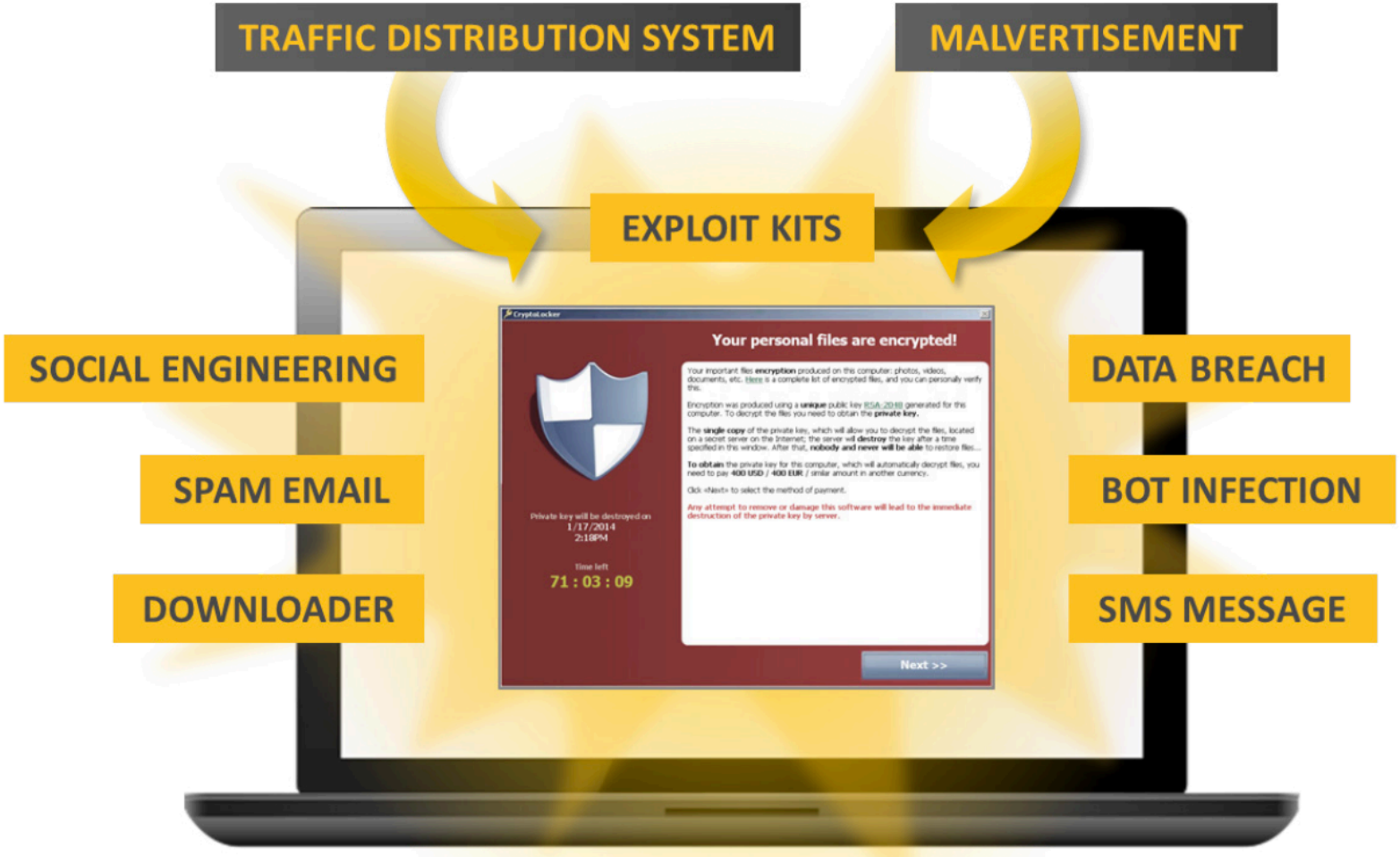


Figure 6: Indexed growth in total number of observed ransomware strains, December 2015-March 2017

Source: Proofpoint Q1 2017 Quarterly Threat Report

- A piece of special malware that infects a computer and restricts access to the computer and/or its files
- Ask for a ransom to be paid in order for the restriction to be removed

Ransomware Propagation



WannaCry

https://www.theregister.co.uk/2017/05/13/wannacrypt_ransomware_worm/

DATA CENTER SOFTWARE **SECURITY** DEVOPS BUSINESS PERSONAL TECH

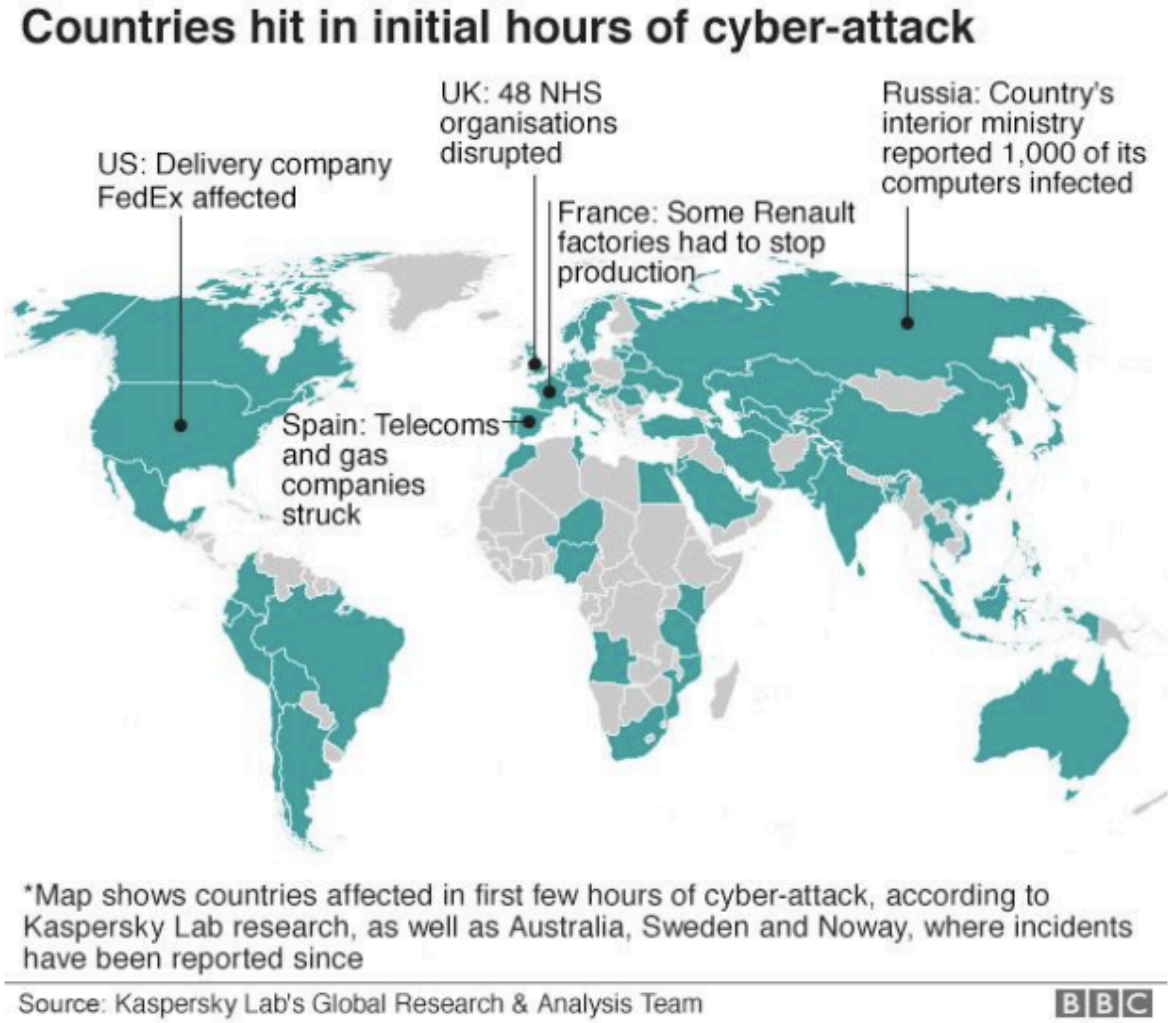
Security

74 countries hit by NSA-powered WannaCrypt ransomware backdoor: Emergency fixes emitted by Microsoft for WinXP+

All you need to know – from ports to samples

By Iain Thomson in San Francisco 13 May 2017 at 00:16 413 SHARE ▼

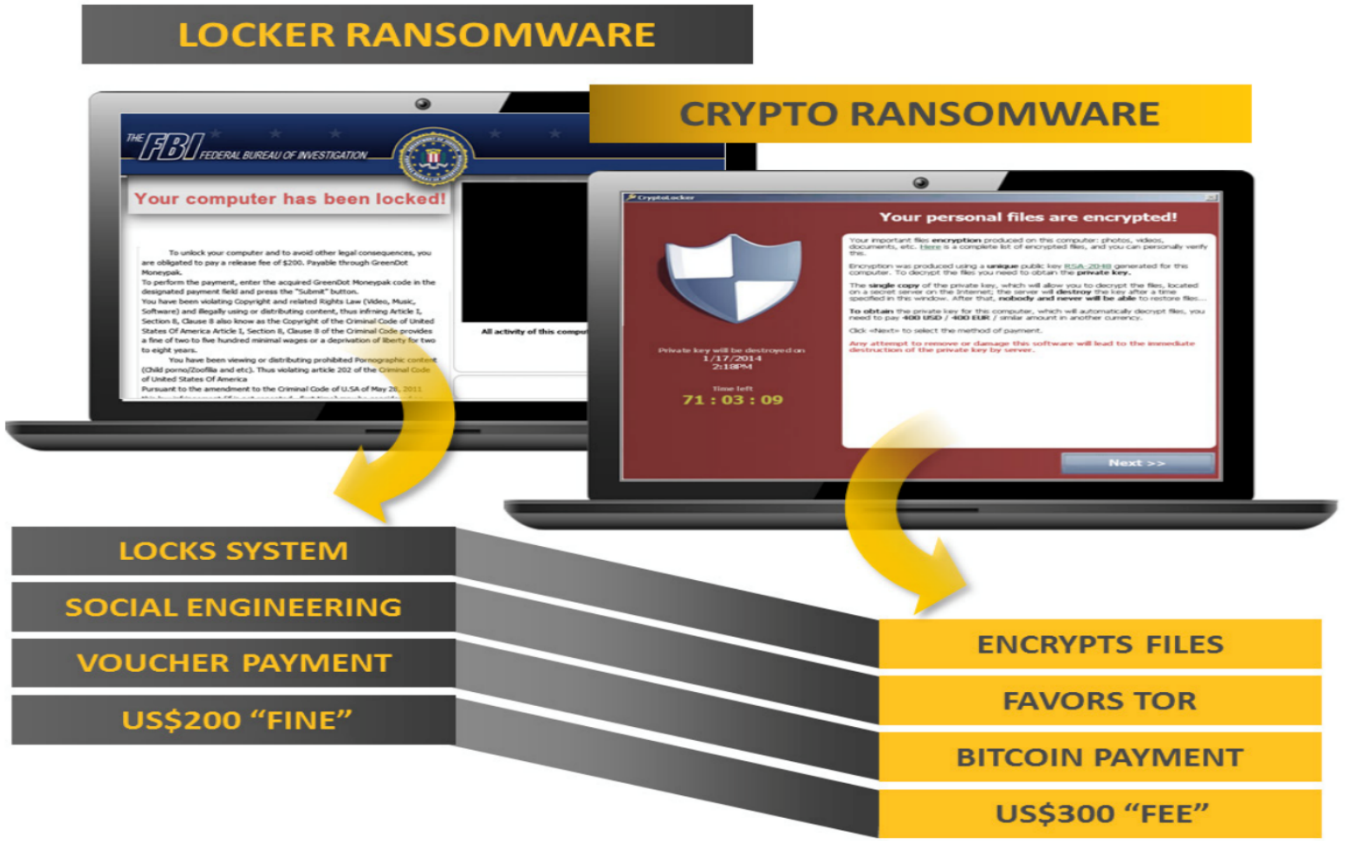




Within a day the code was reported to have infected more than 230,000 computers in over 150 countries

Types of Ransomware

- Locker ransomware
- Crypto ransomware



How to Defend against Locker Ransomware?

- Observation: only the system is locked by the ransomware, but the data are stored intact

- Unplug the storage medium(e.g., hard drives, SSD drives, SD cards), plug the storage medium to a new computing device, and copy out the data

- Plug the storage device back to the device which has been locked, and re-install/initialize the system, then copy the data back

How to Defend against Crypto Ransomware?

- Option 1: detect crypto ransomware **before it causes significant damage to data**
  - Crypto ransomware may be detected since it behaves differently from normal software and other types of malware
  - Crypto ransomware usually encrypts a large amount of data in a short time, and over-writes the old data
    - A large number of read access
    - Expensive computation is required for a large amount of encryptions
    - A large number of writes/over-writes in a short time

The most challenging issue is how to detect the crypto ransomware fastly, since the detection is time-sensitive

How to Defend against Crypto Ransomware (cont.)?

- Option 2: make sure the data encrypted by crypto ransomware are **always recoverable**
  - Create backups, but where to store the backups
    - Cloud? Extra cost for purchasing resources and maintaining the backups
    - Local? How to make sure that the crypto ransomware cannot have access to the backups and then corrupt the backups?

How to Defend against Crypto Ransomware (cont.)?

- Option 3: detection + recovery

Paper Presentation

- ShieldFS: A Self-healing, Ransomware-aware Filesystem

- Presented by Ryan Olson